

David Aronowitz

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EDUCATION

The Open University of Israel

Expected Graduation October 2027

B.Sc. in Computer Science | GPA: 92

EXPERIENCE & EXTRACURRICULARS

First Robotics Competition (FRC), Team Excalibur, high school

2022 - 2024

Software Team Member | Java, Git

- Developed code as part of a 12-person software team, working closely with fellow developers to share tasks, test and build reliable code within a 40-member robotics group (including mechanical and hardware teams).
- Wrote **Java** code to control the robot, implementing **motor control** and **image recognition** systems.
- Participated in **Git** code reviews and meetings, reading teammates' code to catch bugs and maintain a stable codebase before competitions.
- Explained software constraints and features to the hardware builders to align the programming with the physical robot design.

PROJECTS

Stock Trading Simulator & Portfolio Tracker | *Python, Flask, PostgreSQL, AWS, Docker*

- Developed a full stack platform using **Python** and **PostgreSQL** on **AWS** to store and process simulated stock trades, allowing users to practice investing without risking real money.
- Built the frontend using **JavaScript**, **HTML**, and **CSS** integrated via a **Flask RESTful API** to display live portfolio updates so users can immediately see the financial impact of their trades.
- Packaged the app with **Docker** for quick setup and used **AWS Lambda** to automatically record users' daily closing balances, keeping historical account data accurate even when offline.

Custom Assembler | *C*

Open University – Systems Programming Lab

- Implemented a two-pass assembler in **C** to parse, validate, and translate custom assembly code into executable binary instructions, ensuring structural correctness before execution.
- Managed dynamic memory allocation and efficient symbol table structures to prevent memory leaks and handle complex label resolutions reliably.

RESEARCH PROGRAM

Alpha Program

Hebrew University of Jerusalem

Selective national research program for gifted high-achieving students

- Conducted a research thesis on quantum chaos under the supervision of Dr. Noam Hai, actively incorporating expert feedback to refine methodologies.
- Simulated a black hole potential model using **Python**, **NumPy**, and **SciPy** to solve differential equations, investigating unmapped chaotic behaviors to deepen insights into the intersection of quantum mechanics and chaos.
- Analyzed simulation data using **Matplotlib** to compare system behavior against theoretical models, validating physical assumptions and deepening insights into chaotic systems.
- Presented the research at the national Young Developers and Scientists competition.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript, HTML, CSS.

Libraries/Tools: NumPy, Matplotlib, Git, NEAT, Pandas, Docker, AWS, PostgreSQL, SQLite, RESTful APIs, AI, Linux.

Concepts: Data Structures, Algorithms, Object Oriented Programming (OOP), Operating Systems, Computer Organization, System Design, Web Application Development, Software Testing, Networking and protocols, ML.